SoyMate: data report

Photo of a grain sample at intake → separate each grain → classify (sound, broken, insect-holed, stained, foreign material) → % per class → human food or animal feed? Pre-event data exploration, 30/10/2026: no model was trained or run. Figures and sources: CONTEXT.md.

In one line

Feasible as a 20-hour computer-vision project. Good public data exists to train each piece (separating grains, detecting broken grains, detecting holes), but none of it is French pulses in phone photos: we need our own photos taken at the event. Inarix (France) already does this for cereals; we found no one doing it for pulses. A photo only sees visible damage, not the larva inside the grain.

Why it matters

- At intake, someone looks at a sample and decides whether the lot goes to human food (better price) or animal feed. It is visual and manual: LABOSEM (SEMAE) counts bruchid-damaged grains on 200 grains of lentil.
- Pea: only 55 % of 2025 lots had less than 1 % broken grains. Lentil: in the 2025 survey, lots had on average 24.4 % bruchid-damaged grains (40 samples, from 1 % to 75.5 %).
- French contracts (Incograin, 2003 text): at most 1 % bruchid-damaged grains for human food; for animal feed no specific limit (10 % for broken, bruchid-damaged and husked combined). IGP Lentilles vertes du Berry: under 0.1 % unacceptable grains.
- Own calculation: with the indicative prices of the Terres Inovia 2025 guide (280 €/t human food vs 250 €/t feed), a 28 t truck of peas wrongly sent to feed is about 840 €.

The data

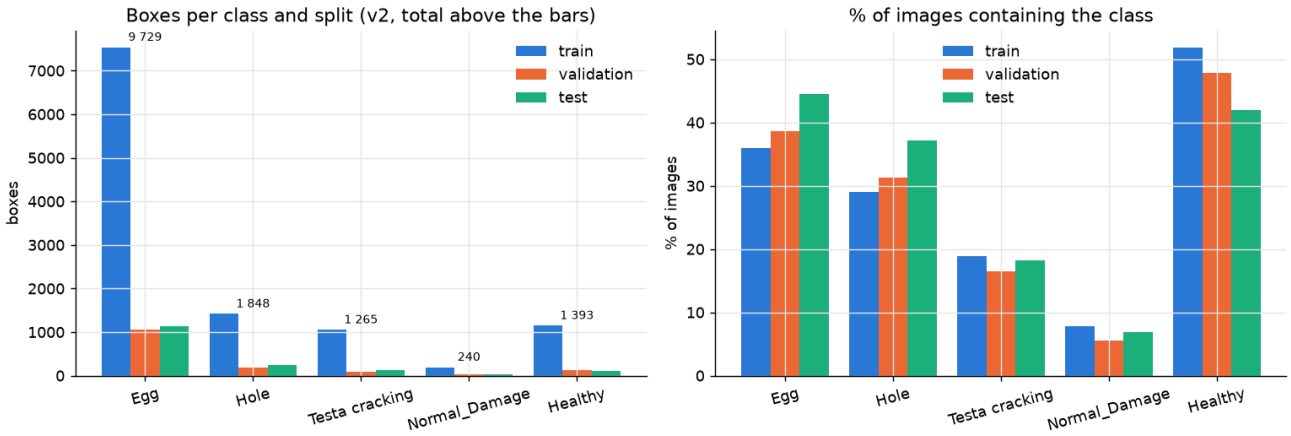
Dataset	Content	Licence	Use
Soybean pulse beetle (Mendeley d97rjyk2tz)	2,726 phone photos, 1 seed per photo, boxes: egg, hole, crack, healthy	CC BY 4.0	Detect insect exit holes
GrainSet (Scientific Data 2023)	350,000 cereal kernels with expert labels: sound, broken, pest-damaged, mould, foreign material	CC BY 4.0	Broken and holes; best label quality
corn1000 (Hugging Face)	836 phone photos with 170-1,000 grains on A4 and exact counts	CC BY-NC 4.0	Separate and count grains in multi-grain photos
Radio_lentille (Terres Inovia)	807 X-rays of bruchid-damaged French lentils + counts	MIT	French validation; request access on Hugging Face
Segmented pea (Zenodo)	5 photos with 2,155 annotated peas	CC BY 4.0	Test grain separation on peas
Faba AAFC (Zenodo)	78 photos of 3-10 faba beans, no damage	CC BY 4.0	Test photos

We found no public image dataset of damaged peas, lentils, faba beans or chickpeas. Plan: train on soybean and cereals, and transfer to pulses with our own photos.

The soybean dataset in depth

1. What is inside

One seed per photo (97.8 %). Eggs dominate the boxes (9,729) because they are tiny and many per seed; hole 1,848, crack 1,265, healthy 1,393.



Boxes per class and split. (research/soybean_bruchid/01_class_counts.png)

2. The exit hole: the class that transfers

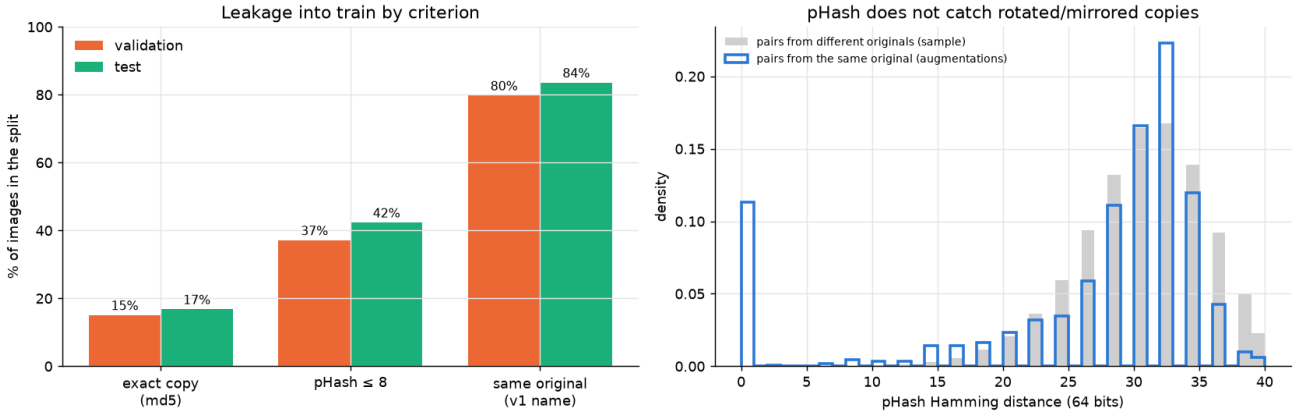
The round hole with a "window" resembles the bruchid hole in faba bean and lentil. Eggs do not transfer: the French bruchid lays its eggs on the pod, not on the grain.



Examples of the "Hole" class with their boxes. (research/soybean_bruchid/04_samples_class1_Hole.png)

3. The published split cannot be used for evaluation

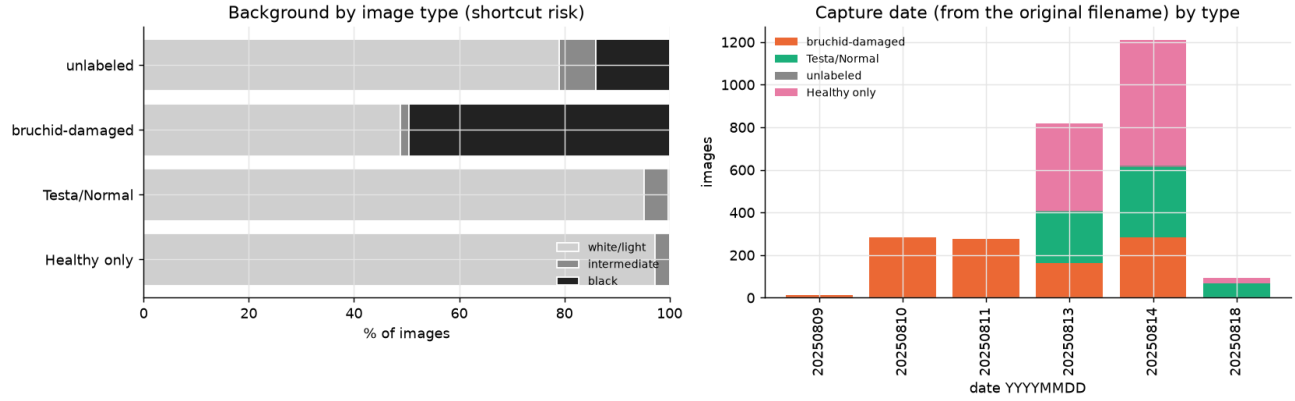
80 % of validation and 84 % of test images are rotations or crops of photos that are also in train. Any metric on that split is inflated. We built a leak-free split by original photo (research/soybean_bruchid/resplit_by_original.csv).



Leakage into train by criterion, and why perceptual hashing misses it. (research/soybean_bruchid/05_leakage_duplicates.png)

4. Shortcuts: background and date give the damage away

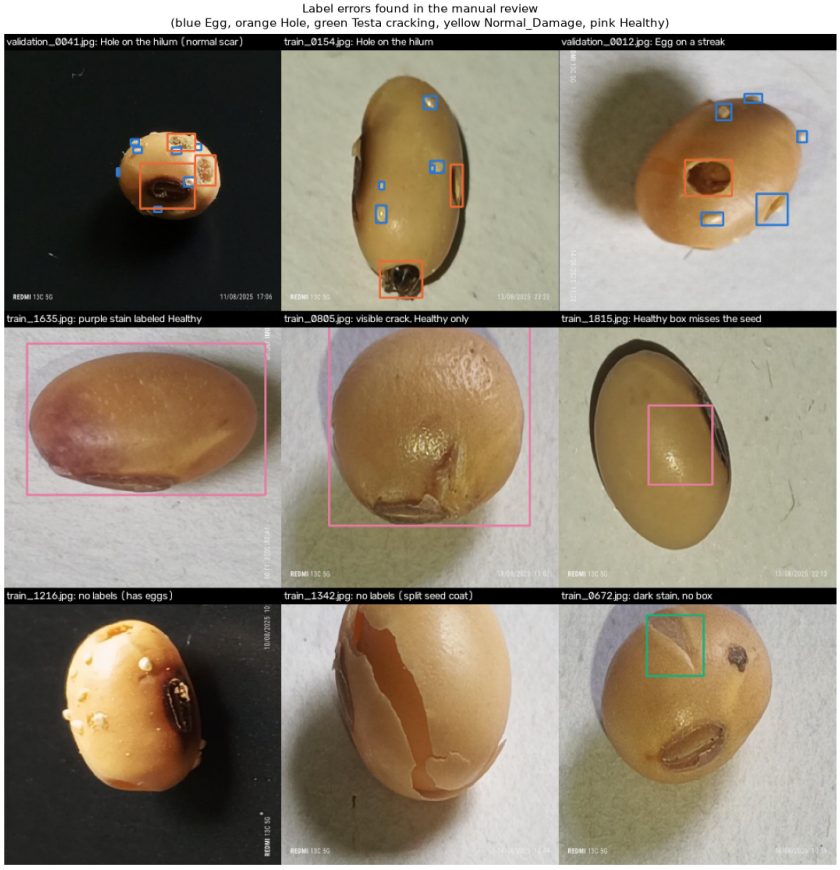
505 of the 513 black-background photos show damage, and the date printed on the photo separates "all damaged" days. A model can learn that instead of the hole: crop the grain before classifying.



Background and capture date by class. (research/soybean_bruchid/03_background_date.png)

5. Label quality: good, not perfect

In 100 photos reviewed by hand, about 15 have a clear error (missing boxes, wrong class, loose boxes). 26 % of the photos with a "Healthy" box also contain defects.



Examples of label errors. (research/soybean_bruchid/07_label_errors.png)

The other datasets

6. GrainSet: the best labels for broken and pest damage

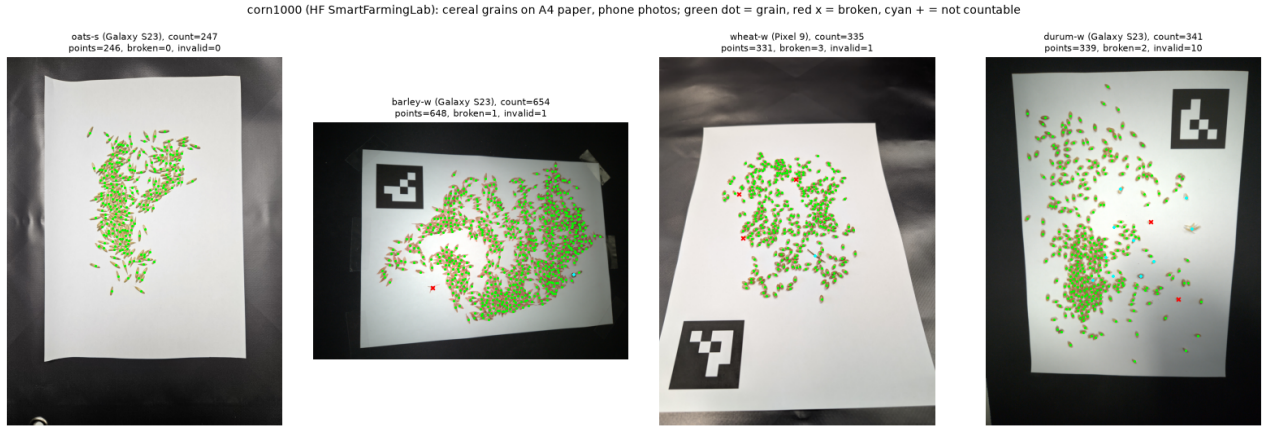
Expert labels per kernel, with very clear "broken" and "attacked by pests" classes. Cereals, but the visual concept transfers.



GrainSet, wheat: one kernel per image, by class. (research/other_datasets/01_grainset_wheat_grid.png)

7. corn1000: the real intake case

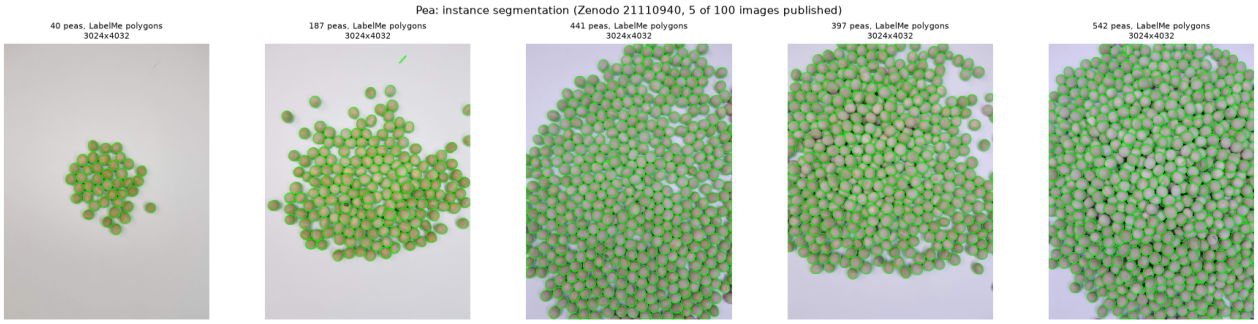
Phone photos of hundreds of grains on an A4 sheet, with exact counts. Used to measure how much grain separation errs.



corn1000: green dots = grain, red crosses = broken. (research/other_datasets/06_corn1000_phone_counting.png)

8. Pea: only for testing grain separation

5 photos with every pea annotated. Enough to check that the method separates touching peas.



Pea instance segmentation (Zenodo 21110940). (research/other_datasets/05_pea_instances_zenodo21110940.png)

Competition and limits

- **Inarix** (France, phone photos): wheat, barley and maize, over 600 silos; in Canada, a project to identify the soybean variety from photos. Nothing found for pulses. EyeFoss: cereals only.
- **Physical limit:** in two pea lots of 200 grains (LABOSEM 2022), the visual count gave 4.1 % and 3.8 %, and dissection 11.8 % and 13.5 %. A photo sees the hole, not the larva inside. X-rays do (the Terres Inovia dataset).
- **Soybean:** on average over 2020/21-2024/25, 45 kt of the 402 kt produced in France went to human food. The pest in the dataset does not seem relevant in France: soybean is the proof of the method, not of the French problem.

How to build it at the event

Step	How	Validated on
1. Separate grains	Threshold and contours on a plain background; pretrained SAM or Cellpose if grains touch (no training)	5 pea photos, corn1000
2. Classify each grain	Features from a pretrained model (DINOv2) + a simple classifier, on CPU. Trained on soybean (leak-free split) + GrainSet + our photos	Leak-free soybean test; own photos split by photo, never by grain
3. Decide the lot	% per class with confidence interval; compare with the threshold (1 %, 3 %)	5-10 mixed photos counted by hand

Photo protocol (~2 h): matte blue or black background, phone fixed at 25-30 cm, diffuse light, 1 € coin for scale; 30-80 grains per photo, not touching; one class per photo; ~50 photos (2 species × 6 classes × 4 photos); label with Label Studio or CVAT; synthetic damage in a separate, declared class.

Before and during Friday

- Request access to **Radio_lentille** on Hugging Face (automatic approval) and download it.
- Buy green and coral lentils, split and whole peas, faba beans, chickpeas and soybeans (Sunday is a holiday; Saturday the campus closes at 18:00).
- Confirm with the organisers: that "Insects" does not exclude detecting insect damage, that soybean for human food is in scope, and that supplies can be bought before the event.
- Contact **Antoine Gardarin** (UMR Agronomie, same campus) for real bruchid-damaged grains.

The pitch message

> "We tested the method on public soybean data that we audited and re-split (the original split leaked 80 %), and transferred it to French lentils and peas with photos taken here. We detect visible damage: holes, broken grains, stains. Inarix does this for cereals; we found no one doing it for pulses, where broken grains and bruchids decide whether a lot goes to human food."

Full detail: research/ (README.md and one findings.md per folder). Source of truth: CONTEXT.md.